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An experimental study of the influence of uniaxial load on flame jet drilling

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ABSTRACT

The performance of flame jet drilling on Grimsel Granodiorite cylindrical rock samples was experimentally investigated depending on applied uniaxial load. With a simple confining system for the rock samples, uniaxial load up to 47 MPa was tested in order to simulate the axial component of the stress tensor existing in a deep borehole. Provided that flame properties were unchanged, increasing the uniaxial load influenced the drilling positively. The experimental results were explained with numerically calculated stress profiles. It was concluded that tensile stresses below the heated surface increase with uniaxial load and additionally weaken the rock, which should improve the flame jet drilling with depth.

1. Introduction

Geothermal energy is an alternative energy resources stored deep in the Earth. It can be harvested producing hot ground water stored in aquifers, or by means of an artificial water circuit sweeping hot and usually hard crystalline rock. However, building of geothermal power plants requires drilling of geothermal wells, which accounts for up to 50 % of the capital costs.[1] A technology called flame jet drilling has the potential to reduce those costs. It is a contactless drilling method characterized by limited drill bit wear[2] and large penetration rates, up to 30 m/h.[3] Therefore, the Institute of Process Engineering at the ETH Zurich works on developing the flame jet drilling into an applicable and competitive drilling technology. The technology can be optimized in terms of drilling velocity, cost-effectiveness and applicability relative to different rock types.

Flame jet drilling is actually impinging the rock surface with a hot gas stream and breaking the impinged spot due to induced thermal stresses in the material. It results into fracturing and creation of thin chips (flakes) i.e. spalls,[4] thus the phenomenon is called thermal spallation. The spalls are being detached and removed with the flow of exhaust gases exposing “fresh” rock layer to the same flame jet. This periodical creation and removal of spalls results in continuous rock disintegration and cavity formation.

So far, it is unknown if stresses acting on a rock mass in nature would influence the generation of thermal stresses during flame jet drilling. Stress tensor acting on a rock mass in nature is generally caused due to gravity and compressive stresses, and it varies from point to point.[5] Gravity causes axial stresses, thus it is expected that flame jet drilling would be influenced when drilling deeper in the field. In order to study the impact of axial stresses on flame jet drilling in laboratory, an uniaxial load was applied on rock samples. It was generated by a simple confinement consisting of two flanges positioned on both ends of the cylindrical rock probe (essentially forming a sandwich structure) and tightened with bolts and nuts. Up to 47 MPa of axial confining pressure was applied, which would correspond to the conditions in a depth of 2–3 km, when drilled through crystalline rock layers only.[6] The influence of uniaxial load on flame jet drilling has never been experimentally studied before.

An additional advantage of the axial confinement was blockage of the sample expansion which resulted in axial breakage (mode I crack) as shown in Fig. 1 (left-hand side). The breakage occurred due to thermal stresses acting in the core of the sample in less than 10 s after exposing the whole top surface to the flame jet. The thermal stresses required for spallation were released through the created fracture and thermal spallation stopped. Similar problems are also described in the literature related to the flame jet drilling technology.[7] With axial confinement, the applied load counteracted the generated stresses and mode I crack did not occur. As a result, the drilling lasted much longer creating large cavities and the drilling conditions in laboratory got closer to field conditions (Fig. 1, right-hand side).

The experimental results can be better understood if the developed thermal stresses, at the heated surface and below it, are known. A non-intrusive method to measure the developed stresses before and during spallation could not be found. Therefore, the generated stress field was calculated and analyzed with a numerical model.

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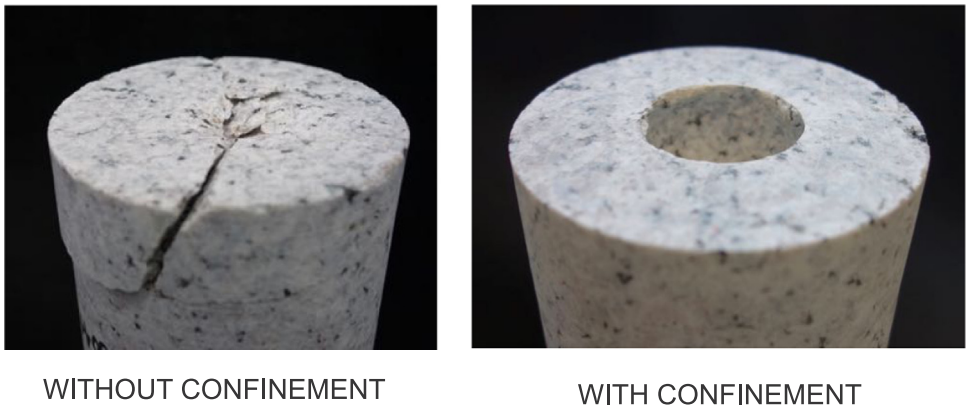


Fig. 1. Rock samples treated with the flame jet. The one on the lefthand side was drilled without and the one on the righthand side with an axial confinement.

2. Current theory of thermal spallation mechanism

If a brittle solid body of low thermal conductivity is suddenly exposed to a high temperature (hot thermal shock) a steep temperature gradient is created inside of it. As a result, the difference in expansion of the body's elements will induce thermal stresses, a fracturing and enlargement of the fracture, which can lead to a breakage. In general, the developed stresses are both tensile and compressive. The region of compressive stresses is a very thin zone at the surface exposed to the heat. The surface layers intend to expand in all directions but the cold layers prevent the expansion causing the compression of the heated elements. Beneath it there is region of tensile stresses because the expanding hot layers pull the deeper cold layers which are as a result under tension.

In thermal spallation the same phenomena occurs but on microscopic scale and very locally as presented graphically in Fig. 2. After exposure of a hard crystalline rock to an impinging flame jet thermal stresses arise due to difference in expansion of grains. Also, rock is full of randomly oriented microcracks and flaws[4] that initiate crack propagation in the compressive region as explained by Preston and White[8] (see Fig. 2, step 1 and 2). Compressive stress does not break the material directly. Under compression, tensile stresses arise that actually propagate rock fracturing and its failure.[8] In addition, the weaknesses can interact with each other and grow (Fig. 2, step 3) as experimentally shown on thin plates by Nemat-Nesser and Hori.[9] They proved that a pre-existing crack in a brittle material extends in the direction of the applied compressive stress. In case of a flaw being not parallel with the surface a shear stress induces the tension at both ends of a crack causing its propagation in alignment to the stress field. According to Rauenzahn the same phenomenon is present in thermal spallation forming a near surface crack parallel to the surface plane (see Fig. 2, step 3).[4] When the crack reaches a critical size, buckling of the incipient spall takes place (Fig. 2, step 4). Once the connection between the spall and the core material is broken, the spall is violently expelled from the surface and entrained with the flow of the combustion exhaust gases (Fig. 2, step 5).

3. Experimental setup

The flame jet drilling setup consisted of two main parts: a burner and a rock sample. The burner was positioned above the rock sample impinged by the flame jet from the top. A schematic overview of the experimental system is presented in Fig. 3. Supply gases were methane

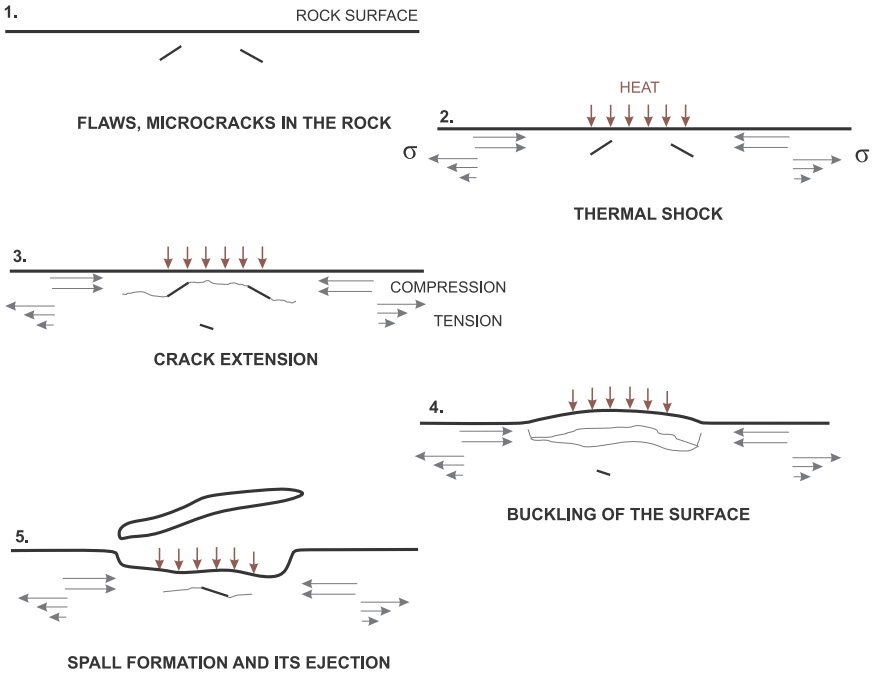


Fig. 2. Schematic overview of thermal spallation mechanism.[10,4,11] Under thermal stresses randomly oriented microcracks and flaws (1, 2) combine with each other and grow (3, 4) until a spall is created and ejected from the surface (5).

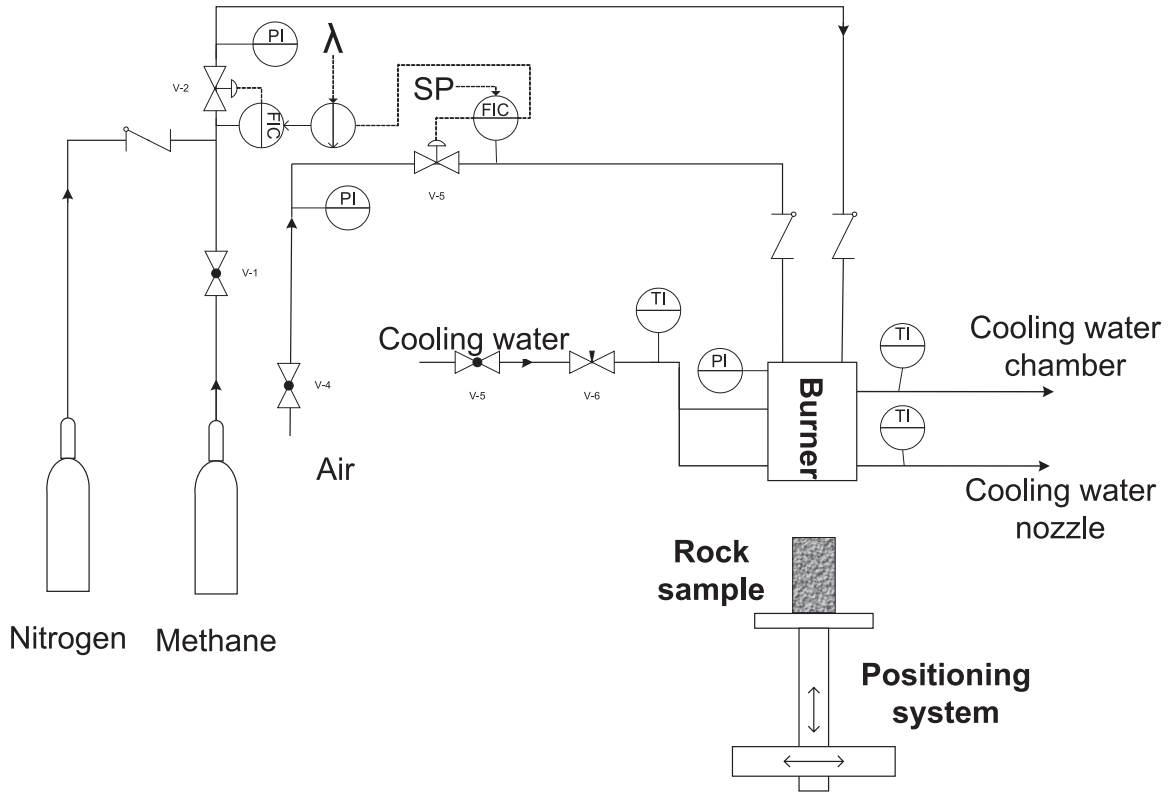


Fig. 3. Schematic overview of the flame jet drilling setup. Methane was used as a fuel and air as an oxidizer. The flame impinged the rock surface from the top. The burner was cooled with water.

as a fuel and air as na oxidizer. The methane gas line was purged with nitrogen after every experiment for the safety reasons. Flow rate of gases was controlled with flow controllers by giving a setpoint (SP) only to the controller of the methane flow rate. The flow rate of air was automatically adjusted according to the chosen air-fuel equivalence ratio (λ), which was 1.1 in all experiments. The cooling water was applied to cool down the burner and protect it from overheating.

The flame power (firing rate) was 38 kW. Hot exhaust gases were exiting the burner through an exit nozzle with 7.9 mm in diameter. The pressure in the combustion chamber was 3.25 bar at chosen conditions. At these conditions, the free flame jet was underexpanded supersonic (eight shock diamonds as a free jet) and turbulent. The temperature at the nozzle (T_{nozzle}) was 1866 K measured with two methods described by Baukal[12]: (1) by means of K-type thermocouples (thickness of 1, 2 and 3 mm) and (2) by a bare S-type thermocouple coated with Ceramabond 569 as described by Taylor et al.[13]. In case the S-type thermocouple is used, Siewert[14] suggested the correction of the recorded temperature, which was implemented in this study. The final reported flame temperature was the mean value of the results obtained with both methods.

The rock sample was positioned below the flame jet with a two axis system (for axial and lateral repositioning). In all experiments the axial distance between the nozzle and the rock surface was 63.2 mm, which was the smallest feasible distance.

3.1. Tested rock samples

The rock samples used in the experiments had cylindrical shape 100 mm in diameter and 150 mm in length. The raw Grimsel Granodiorite was obtained from a quarry in Chuenzentennlen close to Grimselpass in Switzerland. It is possible that this type of rock will be drilled in deep geothermal project in Switzerland and it is well characterized. The rock consists of quartz, K-feldspar, plagioclase and biotite. The physical properties of the rock are density 2706 kg/m^3, heat capacity 1025 J/(kgK), Young's modulus 47.3 GPa, Poisson's ratio 0.33, thermal expansion coefficient $8.5 \times 10^{-6}\ 1\backslash K$ and thermal conductivity 3.22 W/(mK).[15,16]

3.2. Applied uniaxial pressure

The rock samples were pressed uniaxially with two flanges and tightened with four bolts as shown in Fig. 4. In order to provide a more uniform distribution of the pressure along the surface, an additional disc made of steel was inserted between the upper flange and the rock sample. Without the disc, mostly the edge of the rock sample would be pressed due to possible elastic deformation of the flange while tightening the bolts. In order to be able to use the flanges many times they were constructed of Toolox33, which does not deform plastically after applying the confining pressure.

Following the manufacturer recommendation, the maximal torque applied on the nut of each bolt was 180 Nm. A load cell (Induk GmbH, HKMD43LT-G1) was used to measure the resulting force on one bolt. In the centre of the cylindrical load cell the bolt was placed with one nut above and one under the load cell. The applied force on the upper nut was transferred to a disc and from the disc to the load cell. The cylindrical discs were installed in order to distribute the pressure of the nut to the surface of the load cell evenly.

In order to convert the measured force into pressure acting on the rock sample the following equation was used:

$$P_{conf} = \frac{4F_{bolt}}{A_{rock}} \tag{1}$$

where P_{conf} is the applied uniaxial load in MPa, F_{bolt} is the force applied on the rock with one bolt and A_{rock} is the loaded surface of the rock.

The influence of the uniaxial pressure on the drilling was studied for a torque (τ) equal to 0, 60, 100, 140 and 180 Nm. A torque equal to 0 Nm means that no force was applied on the bolts. In this case, the only force acting on the rock was the weight of the flange.

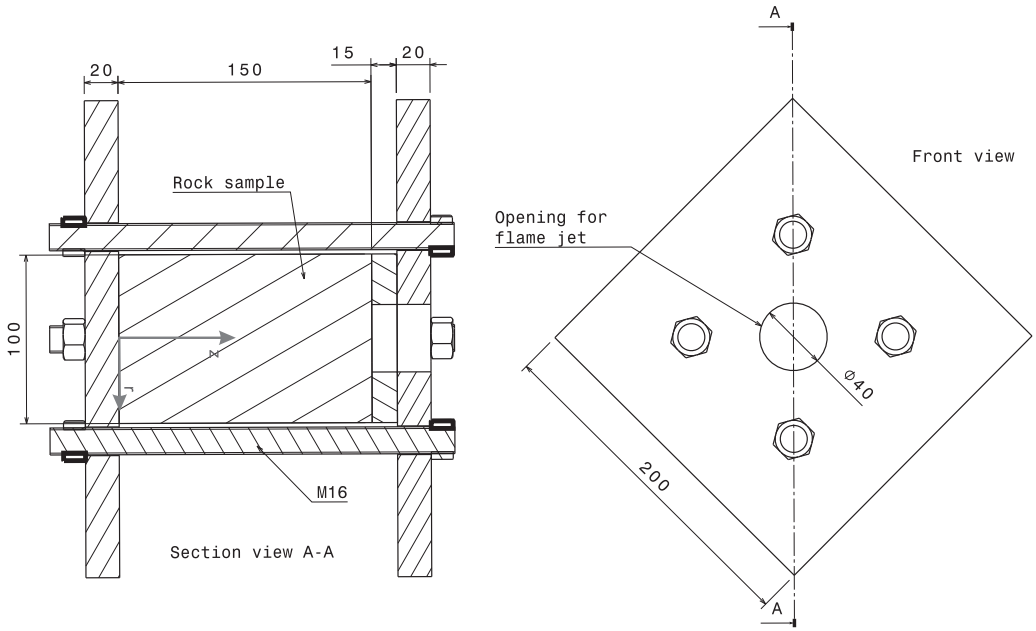


Fig. 4. Schematic overview of the confined rock sample.

3.3. Experimental procedure

When the desired flame conditions had been reached, the confined rock sample was repositioned first upward in axial direction closer to the flame jet and then laterally below the flame jet. At the moment when the flame jet impinged the rock sample the time recording started. The position of the rock sample was not modified while drilling. Also, the total drilling time was equal for all experiments. After the drilling time elapsed (t), the rock sample was removed first laterally and then axially from the flame jet. The removed rock volume ($\dot{V}_{removed}$) in cm^3/min was calculated as in Eq. (2):

$$\dot{V}_{removed} = \frac{m_{removed}}{\rho t} \tag{2}$$

where ρ is density of the rock and $m_{removed}$ is the removed rock mass, which is a mass difference of the rock sample before and after the drilling. The measurement was repeated three times for every loading condition.

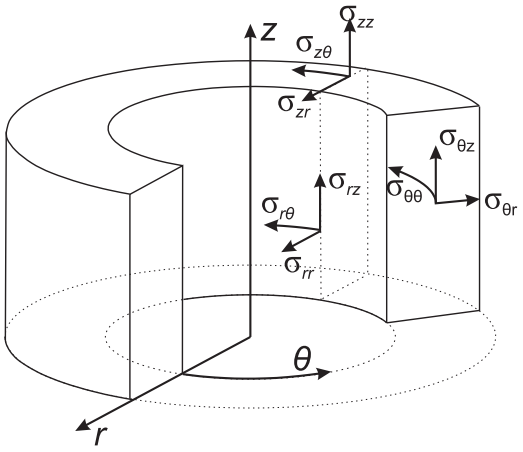


Fig. 5. Normal and shear stresses in cylindrical coordinates.

4. Modelling of stress field in the rock sample

Theoretical analysis of the developed stresses (σ) in the rock sample was performed with a model developed by Meier et al.[17]. It helps to understand the thermal stress development in an observed object when applying a heat source and a load on it. The model is based on uncoupled, quasi-static, thermoelastic equations applied on 2D axisymmetric domain. It is defined with conservation equations for energy and momentum as presented elsewhere.[17] The rock was modelled as a linearly elastic, isotropic, homogeneous, nonporous and brittle medium with the dimensions of the tested rock samples. Required rock properties were density (ρ), heat capacity of the rock sample (C_p), Young's modulus (E), Poisson's ratio (ν), coefficient of thermal expansion (α) and thermal conductivity (k). It was assumed that the properties of the rock are independent of temperature. This assumption does not substantially affect the main conclusions of this paper. The equations are valid for the cylindrical coordinate system presented in Fig. 5.[18] To complete the system of equations the constitutive equation (generalized Hook's law) for the isotropic, linear elastic body is also considered. It presents the stresses-strains relations inside the body.[17]

Initial and boundary conditions based on laws of energy and momentum conservation are graphically illustrated in Fig. 6. The initial temperature of the body is 298.15 K. Convective heat transfer is applied at the top surface of the rock sample exposed to the flame jet ($0 < r <$ 0.02 m, z= 0.150 m). Transferred heat flux ($\dot{q}$) is defined with Eq. (3):

$$\dot{q} = h(T_{ref} - T) \tag{3}$$

where h is the heat transfer coefficient and the T_{ref} is the reference temperature, i.e. flame temperature (1866 K) and T is the surface temperature of the rock. The heat transfer coefficient (h = 7 kW/(m^2K)) was obtained from a separate heat transfer study performed with a heat flux sensor. In the heat transfer study the heat flux to the sensor ($\dot{q}$), its surface temperature (T) and the flame temperature (T_{ref}) were measured and applied in the Eq. (3) as described by Rothenflu et al.[19]. The heat flux sensor was designed to obtain the surface temperatures between 673 and 873 K, which were similar to the temperatures at which the rock sample started to spall, thus it was assumed that the obtained heat transfer coefficient is valid also for the heat transfer between the flame jet and the rock surface. All other boundaries of the domain are defined as adiabatic ($\dot{q} = 0$). Regarding conservation law of momentum, no external stress (traction free) is applied on the boundaries that are not confined (the surface part exposed to the flame jet and curved surface of the rock sample). At the bottom of the rock a "*no − slip*" condition is applied (the body velocity at the boundary is equal to velocity of the

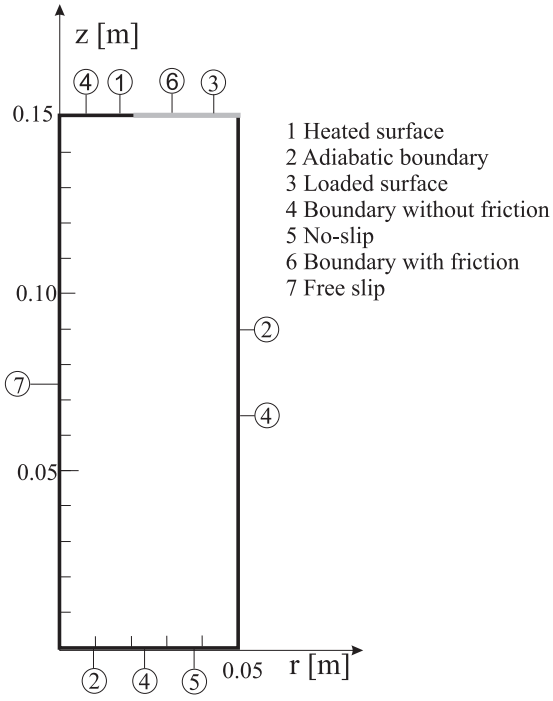


Fig. 6. Schematic overview of the boundary conditions applied in the stress field model.

boundary (zero)). On the upper boundary of the rock (r> 0.02 m and z= 0.150 m), which is pressed with the flange, normal stress (from 0 to 47 MPa) and friction force are imposed. And finally, the symmetry axis is defined as a free slip thus the displacement normal to the wall is zero.

In the experiments at the applied process conditions, the onset of thermal spallation occurred less than 0.5 s after the beginning of the flame jet impingement. The model simulates the development of thermal stresses before the onset of spallation.

5. Results and discussion

In this chapter results of the load cell measurement and the drilling are presented.

5.1. Load cell measurement

The measured normal force increased linearly with increasing torque (τ) on the bolt. The corresponding pressure (from Eq. (1)) on the rock surface is presented in Fig. 7. The results are in agreement with the theory stating that the bolt clamping force is linearly dependent on the tightening torque. In literature, it was shown that the slope of the linear function is dependent on the friction coefficient (nut factor) and the bolt diameter which are both constant with applied torque.[20] The measurement was reproducible and the same results were also obtained with another bolts. When no torque on the bolts was applied only the flange was pressing the rock surface with its weight, which resulted in a pressure of around 9kPa. This is negligible compared to the pressure acting when a torque was applied.

5.2. Flame jet drilling evolving with time

During the flame jet drilling experiments the created cavity in the rock sample was extending in length and diameter with time. The drilling lasted until the diameter of the cavity was almost equal to the diameter of the rock sample. At that moment the thin walls of the cavity broke due to the applied uniaxial load and the drilling could not proceed. For 47 MPa of applied axial load this happened after about 120 s

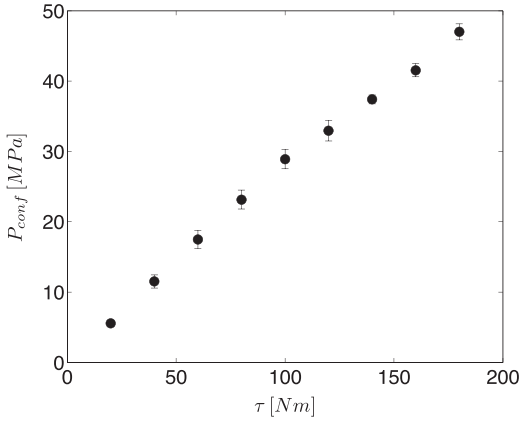


Fig. 7. Uniaxial pressure on the rock sample depending on the torque applied per one bolt.

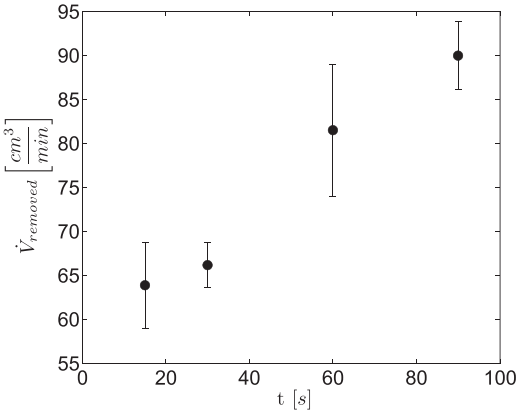


Fig. 8. Influence of time on the removed rock volume when drilling 47 MPa confined rock samples confined the flame jet.

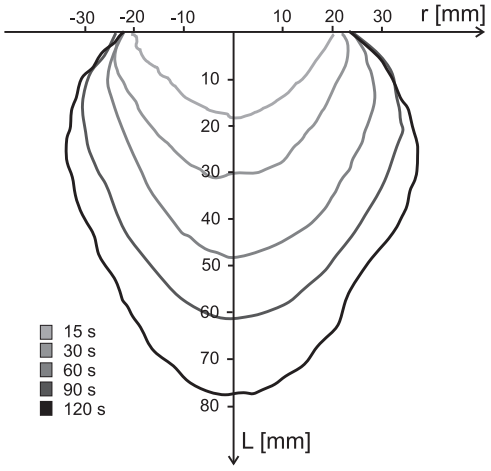


Fig. 9. Propagation of drilled cavity with time. Applied uniaxial load on the rock samples was 47 MPa.

of drilling. The drilling results were evaluated by removal rate ($\dot{V}_{removed}$), which is shown in Fig. 8. It was increasing with time while drilling, which can be seen also in Fig. 9 where the cavity evolving with time is presented, because the rock surface that is exposed to the impinging flame jet is becoming larger, thus more surface is spalling.

Each rock sample that was drilled for a certain period of time was afterward cut to obtain the cross section of created cavity (see Fig. 9). It can be seen, that the cavity grew more in depth than in width with time,

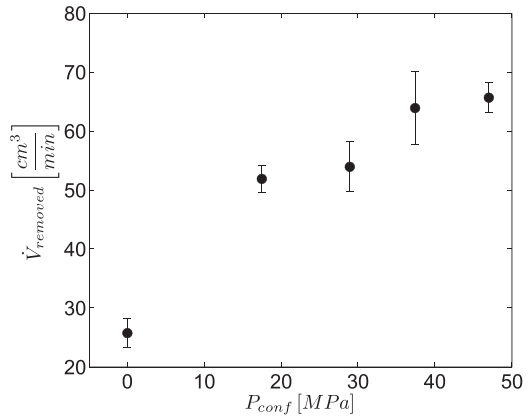


Fig. 10. Influence of uniaxial pressure on removed rock volume while drilling for 30 s.

thus diameter (D) and depth (L) of the cavity are linear dependent with time t where $\frac{dL(t)}{dt} = 1.8\frac{dD(t)}{dt}$). The thermal spallation occurs at the whole rock surface exposed to the flame jet, but the removal is more intense at the stagnation point of the impinging jet. This is probably due to enhanced heat transfer from the flame jet to the rock surface in that region inducing larger compressive stresses. When the rock sample was drilled for longer than 30 s, the cavity was becoming wider than initial impinged free surface. After the period of 30 s the rock sample confined with 0 MPa could be successful drilled without causing its breakage. Therefore, this period was selected for examining the influence of uniaxial load on the flame jet drilling results. After 30 s the standard deviation of the removal rate was 5%, which indicated that the results are reproducible even though not much rock volume was removed.

5.3. Flame jet drilling evolving with pressure

Fig. 10 shows that the removal rate increases with the applied load for the tested loading range (0– 47 MPa). This indicates that the efficiency of the flame jet drilling would increase with depth while creating a borehole, which presents an advantage of this technology over the conventional ones. The tensile strength of the rock is reached much faster with larger uniaxial pressure at the same firing rate. Ejection of spalls is a very fast process (faster than 1/10 000 s as observed with a high speed camera), thus the steps of thermal spallation mechanism (crack extension and buckling) are difficult to detect experimentally. Therefore, a theoretical explanation for the increase of the drilling efficiency with uniaxial load was found in the numerical results and it is further discussed in Section 5.4.

5.4. Stress analysis

According to the model, the uniaxial load and the heat source induce stresses in the rock probe creating regions of compression and tension. Fig. 11 presents the normal stresses in radial (r) and axial (z) directions (σ_{rr}, σ_{zz}, respectively) with 47 MPa of applied uniaxial load and after 0.01 s of heat exposure. Negative values stay for for compression, and positive ones for tension. On the left-hand side of Fig. 11 a narrow region below the surface (including the surface itself) represents compression with a very steep gradient along z - axis. Below the compressive region there is a larger region of tension with lower gradient of stresses. The obtained stress distribution is in agreement with the numerical results from Germanovich,[10] who observed only the influence of heat on stress development. The right-hand side of Fig. 11 shows the stresses in axial direction. The region below the loaded surface ($0.02 < r < 0.05$ m) is exposed to axial compression and due to the elastic properties of the body the compression is transferred to the rock core. Below the free surface (5 mm thickness) the axial stresses are equal to zero because no uniaxial loading is applied.

The influence of time at constant load (47 MPa) and the influence of uniaxial load at the same moment (0.01 s) on development of normal stresses in axial (z), radial (r) and tangential (θ) directions (σ_{zz}, σ_{rr}, $\sigma_{\theta\theta}$, respectively) along the centerline of the modelled rock sample are presented in Fig. 12 and 13, respectively. The intensity of the stresses only close to the impinged surface (free surface) (between 0.1 and 0.15 m) is shown. This is the region where the radial and the hoop stress are mainly influenced by the uniaxial load and the applied heat.

Fig. 12 shows that with time, the heat penetrates deeper in the body, thus the compressive region becomes larger and the tensile region more intense (σ_{rr}, $\sigma_{\theta\theta}$). Fig. 13 shows that beneath the surface by increasing the load up to 47 MPa only the radial and hoop tensile stresses (σ_{rr} and $\sigma_{\theta\theta}$) increase. It occurs due to axial displacement of the region beneath the loaded surface, which pulls the elements below the free surface in radial and tangential direction. The radial compressive stresses (σ_{rr}) at the top layer decreases negligible compared to the stresses when no load is applied. Therefore, the improvement of experimental drilling with uniaxial pressure is related to the increase of tensile stresses in tensile region below the surface. The tensile stresses additionally disintegrate the material and weaken the connection between the grains. Reduced connection strength between grains simplifies the breaking process when entering the compressive region during thermal spallation, which increases the drilling efficiency. Previous theories of thermal spallation mostly highlighted the influence of compressive stresses, but the importance of the tensile region was only discussed in the theoretical work done by Meier et al.[17] and is confirmed with the experimental results presented in this work.

The buckling phenomenon of a spall before ejection can be better

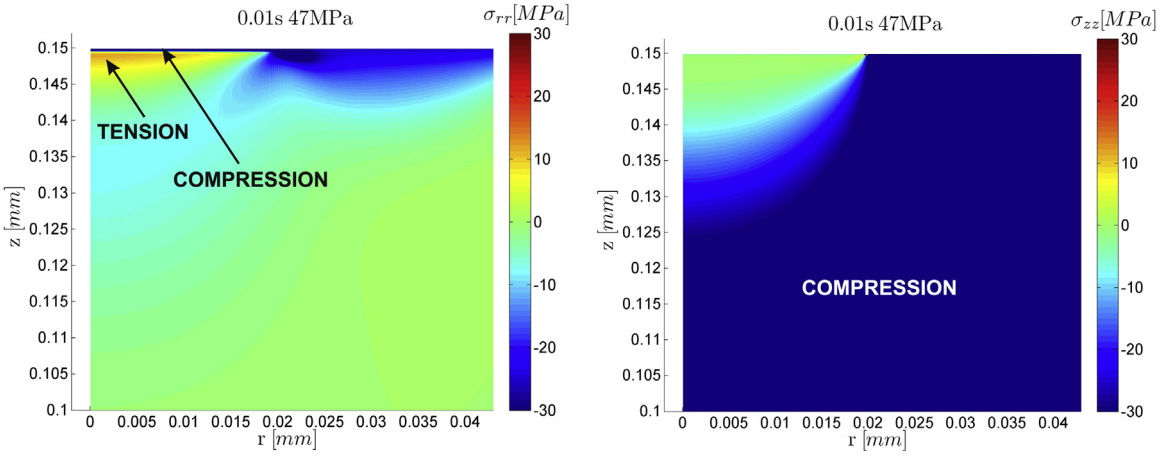


Fig. 11. Stress field in modelled rock in radial and axial direction computed at 0.01 s with 47 MPa uniaxial load.

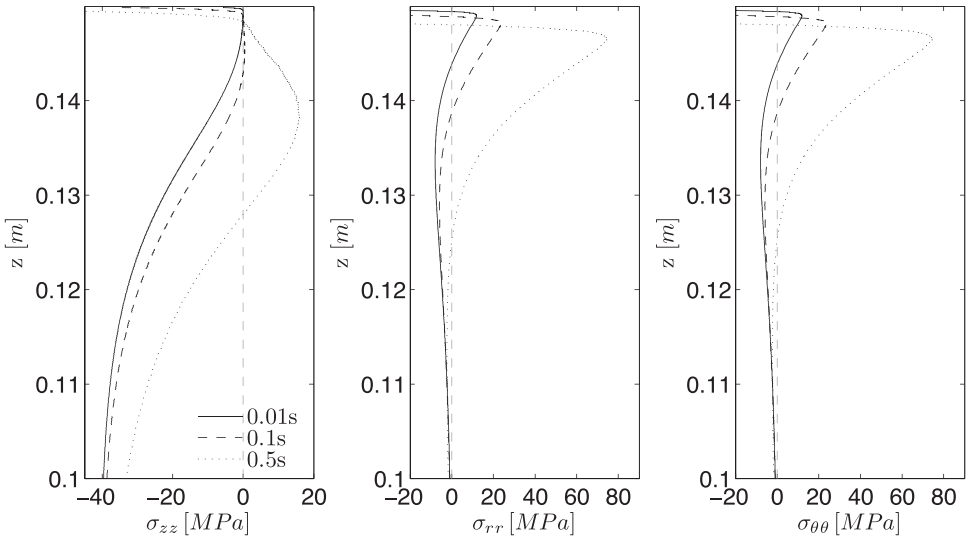


Fig. 12. Stress development with time along the centerline of the modelled rock sample in cylindrical coordinates when 47 MPa uniaxial load is applied.

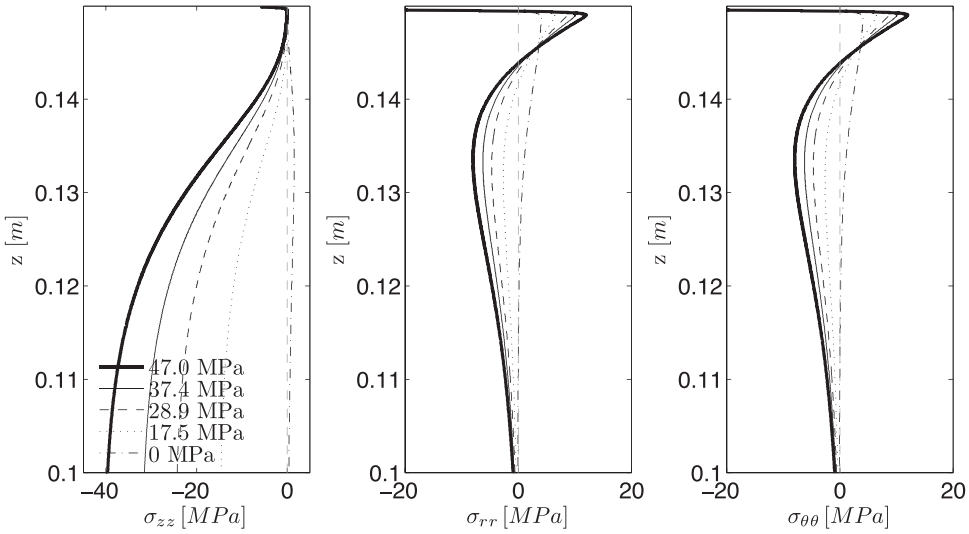


Fig. 13. Stress development with uniaxial load along the centerline of the modelled rock sample in cylindrical coordinates computed at 0.01 s.

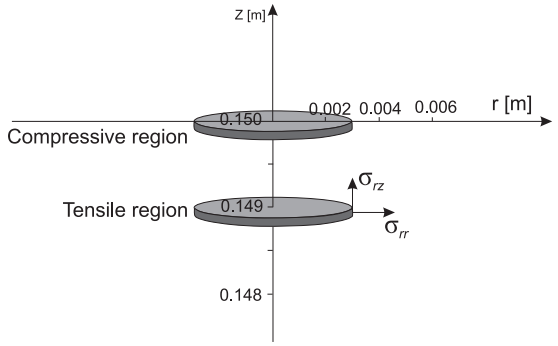


Fig. 14. A spall presented with a pad while placed in the rock sample (in the regions with the largest compressive and tensile stresses). The average stresses calculated as in Eq. (4) and (5) are valid for the curved surface of the pad.

presented by stresses that act on the incipient spall when it is not yet detached from the rock. This spall is located at the surface in the compressive region. It was approximated with a circular pad that simulates the future spall (250 μm thickness and 3 mm in diameter). Stresses that cause crack formation and buckling act along the curved surface of the pad, thus they were additionally analized. The same stresses were observed when the pad is placed in the tensile region (as shown in Fig. 14), actually at the location where the tension is the largest. The acting stresses are interpreted with average stresses ($\overline{\sigma}_{rz}$ and $\overline{\sigma}_{rr}$) calculated as in Eq. (4) and (5), respectively. Those were the stresses of the curved pad surface with remarkable large values compared to the stresses acting on other surfaces of the pad.

$$\overline{\sigma}_{rz} = \frac{\int\int r\sigma_{rz}\,dz\,d\theta}{\int\int r\,dz\,d\theta} \tag{4}$$

$$\overline{\sigma}_{rr} = \frac{\int\int r\sigma_{rr}\,dz\,d\theta}{\int\int r\,dz\,d\theta} \tag{5}$$

The resulting average stress along the curved surface of the pad, depending on its position (compressive or tensile region), are presented in Fig. 15. In the compressive region considerably high stresses were only in radial direction, which decrease with the load. This can not be the cause for the improved drilling, because those stresses are required for the thermal spallation. Contrary to the compressive region, the average tensile stresses acting in radial and axial direction ($\overline{\sigma}_{rr}$ and $\overline{\sigma}_{rz}$, respectively) increase both with the load, which again indicate that the tension influences the drilling efficiency when increasing the uniaxial

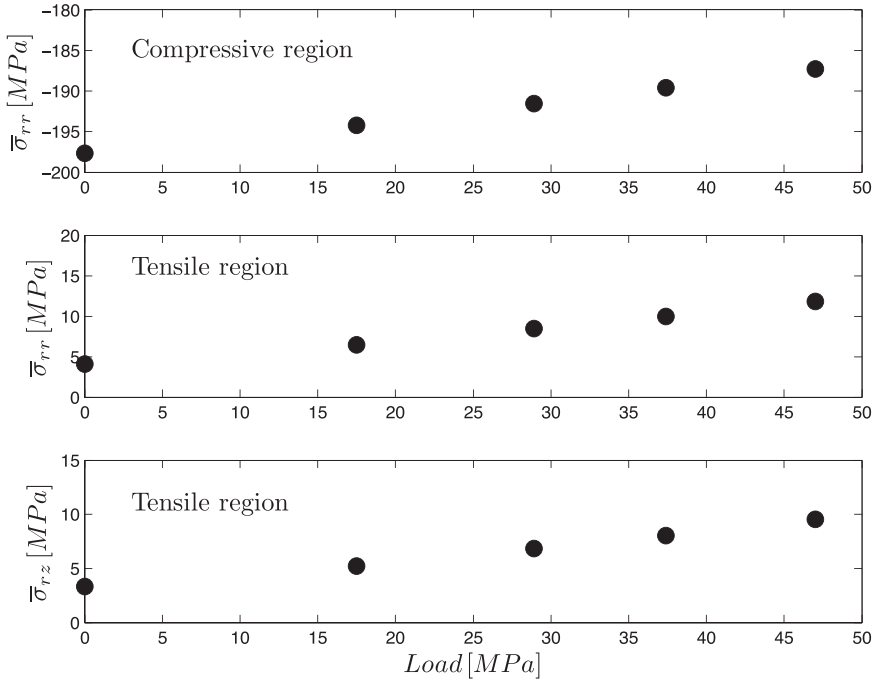


Fig. 15. Computed average stresses acting on a curved surface of a circular pad (250 μm thickness and 3 mm in diameter) when placed in the region of the largest compression and tension in the rock sample.

load as mentioned before.

6. Conclusion

The influence of the uniaxial load on flame jet drilling has been studied. A load of up to 47 MPa was applied uniaxially on a rock sample of Grimsel Granodiorite. The experimental results showed that larger uniaxial load improved the removal rate at the same firing rate. According to the stress analysis an increase of the uniaxial load mainly increases the tensile stresses beneath the impinged surface, which is, thus, related to the improved penetration. A possible explanation is that the generated tension below the impinged surface helps weakening the connection between the grains in the rock sample. As soon as the disintegrated region gets under compression, the spalls creation occurs easier than without the acting uniaxial load. In summary, the new contribution to the thermal spallation mechanism theory is that the tensile stresses below the compressive region help weakening the rock layers additionally to the compressive stresses in the compressive region. This is a beneficial discovery for the flame jet drilling indicating that the rock removal rate should increase with depth being in advantage compared to conventional drilling technologies.

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